RESEARCHLY AI: AI-POWERED RESEARCH ASSISTANT AND COLLABORATION PLATFORM FOR UNIVERSITY STUDENTS

RESEARCH PERFORMANCE ANALYTICS AND VISUALIZATION MODULE: TREND FORECASTING, QUALITY SCORING AND SUCCESS PREDICTION

Individual Thesis

Jayasundara H.W.S.S.

(IT22553096)

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Student name	Student ID	Signature	Date
Jayasundara H.W.S.S.	IT22553096		

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Role and name	Signature	Date
Supervisor: Ms. Sanjeevi Chandrasiri		
Co-supervisor: Ms. Ishara Weerathunga		

ABSTRACT

Universities oversee student research with static spreadsheets and manual rubrics, and bibliometric tools describe past literature without forecasting where an institution's own research is heading. This thesis presents the Research Performance Analytics and Visualization Module (Module 4) of Researchly AI, an integrated research platform built for SLIIT. Using the 4,219-record SLIIT corpus, the module fits per-topic ARIMA(1,1,1) models that forecast publication activity with 95% prediction intervals, trend direction, emergence and recommendations; scores titles and abstracts on four weighted dimensions (originality 30%, citation impact 25%, methodology 25%, clarity 20%) by blending XGBoost regressors with calibrated heuristics; and predicts success probability by blending an XGBoost classifier (40%) with a heuristic (60%). The module follows Design Science Research and is deployed as a FastAPI micro-service with interactive Next.js dashboards.

A hold-out back-test (2024–2025) gave ARIMA a MAPE of 15.3% for the largest series (84.7% accuracy, above the 78% target) but showed it did not beat a naïve forecast for four of six topics. Quality and success models reported $R^2 \approx 0.997$–0.999 and ROC-AUC 0.9994, which the analysis attributes to target leakage from heuristic labels. In practice, scoring discriminated a technical abstract (0.83) from a non-research form (0.24) in under 2 ms. Of 12 validation cases, 7 passed, 1 was partial and 4 failed, all failures stemming from the trend module's partial-year growth baseline and insight text.

Keywords: research analytics, ARIMA forecasting, quality scoring, success prediction, XGBoost

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
AUC	Area Under the (ROC) Curve
BERT	Bidirectional Encoder Representations from Transformers
CI	Confidence (prediction) Interval / Continuous Integration
CPU	Central Processing Unit
DSR	Design Science Research
F1	Harmonic mean of precision and recall
FR	Functional Requirement
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
JWT	JSON Web Token
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
NRMSE	Normalised Root Mean Square Error
PDF	Portable Document Format
PDPA	Personal Data Protection Act (Sri Lanka, No. 9 of 2022)
REST	Representational State Transfer
RLS	Row-Level Security
RMSE	Root Mean Square Error
ROC	Receiver Operating Characteristic
SBERT	Sentence-BERT
SD	Standard Deviation
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
SO	Specific Objective
SUS	System Usability Scale

Abbreviation	Description
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing
UI	User Interface
XGBoost	Extreme Gradient Boosting

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1 INTRODUCTION

1.1 Background and Overall Project Context

University students face a research workflow that has grown far faster than the tools supporting it. Global scholarly output rose from roughly 2.1 million articles in 2010 to more than 3.3 million in 2022 [1], and a final-year undergraduate must survey that literature, select a defensible topic, find a suitable supervisor, cite sources correctly, avoid overlap with earlier work and write a structured proposal, usually within a single academic year and with limited supervision time. Each of these tasks is served by a separate tool — a reference manager, a search engine, a similarity checker, a spreadsheet of supervisors — and none of these tools share data [2], [3].

The group project R26-IT-116, **Researchly AI — an AI-powered research assistant and collaboration platform for university students**, addresses this fragmentation by placing four intelligent modules behind one web application and one shared data substrate: Module 1 (Research Integrity and Compliance), Module 2 (Collaboration and Recommendation), Module 3 (Research Data Collection and Management) and Module 4 (Research Performance Analytics and Visualization). All four modules are grounded in a corpus of 4,219 records drawn from SLIIT's institutional research repository (RDA), of which 3,858 carry usable abstracts, so that every recommendation, gap, forecast or similarity score refers back to real prior SLIIT research rather than to generic web content. The platform is deployed as a Next.js front end on Vercel, an Express.js API gateway and four FastAPI micro-services, backed by Supabase PostgreSQL with the pgvector extension (Figure 1.1).

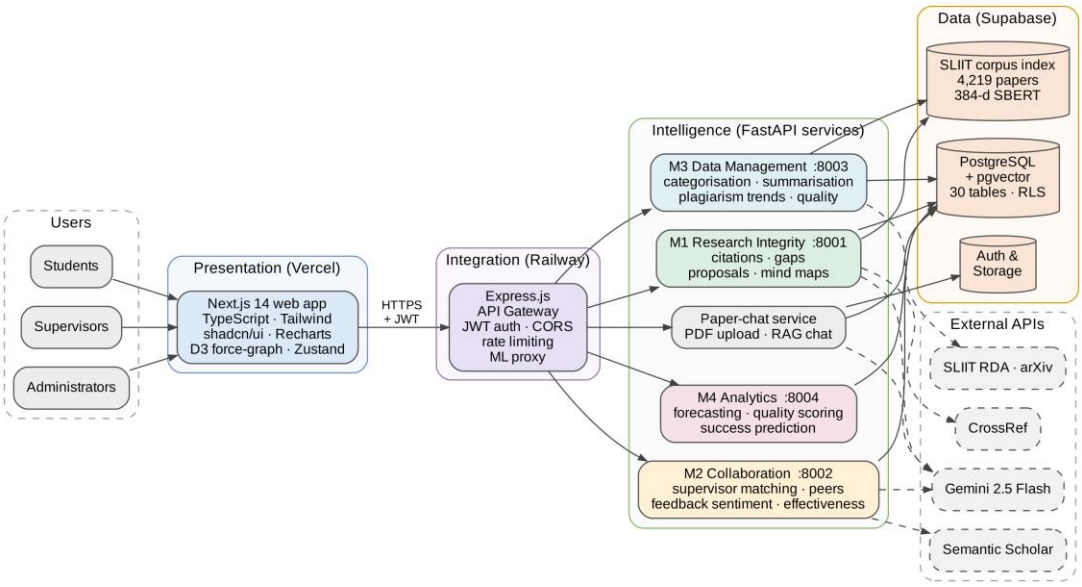


Figure 1.1: High-level architecture of the Researchly AI platform showing the presentation, integration, intelligence and data layers

This thesis reports the author's individual component, **Module 4: Research Performance Analytics and Visualization Module**. The common group report describes the integrated solution, the shared infrastructure and the end-to-end evaluation; this document concentrates on the research problem, design, implementation and validation of the individual component, and on how it exchanges data with the other three modules.

1.2 Individual Component Context

Module 4, the **Research Performance Analytics and Visualization Module**, is the foresight layer of the platform. It turns the corpus maintained by Module 3 into forward-looking, decision-supporting numbers for students, supervisors and administrators. It is implemented as the FastAPI service `services/module4-analytics` (port 8004, all routes under `/analytics`) and three pages under `/analytics`:

- **Trend forecasting** — per-topic publication forecasts with 95% prediction intervals, trend direction, growth, model accuracy, multi-topic comparison, emerging-topic detection and research-direction recommendations.

- **Quality scoring** — a four-dimension score (originality, citation impact, methodology, clarity) for any title and abstract, or an uploaded PDF/TXT file, with predicted discipline and improvement recommendations.
- **Success prediction** — the probability that a paper or proposal meets the platform's definition of a successful research output, with a risk band and targeted recommendations.

1.3 Problem Statement

Universities manage research portfolios with static spreadsheets and manually applied rubrics. Bibliometric tools such as VOSviewer and CiteSpace visualise past literature [4], [5] but do not forecast where an institution's own research is heading. Citation-based quality indicators have long lags that make them meaningless for student work [6], and rubric scoring is slow and inconsistent between assessors. Students therefore choose topics without knowing whether an area is growing or saturated, and supervisors identify weak proposals late. The problem addressed by this component is: **students and institutions lack transparent, timely analytics that forecast research-area trends from the institution's own output and give early, explainable feedback on the quality and likely success of individual research work.**

1.4 Research Gap

Table 1.1: Research gaps addressed by Module 4

ID	Gap	Evidence
G1	Bibliometric tools describe past literature but do not forecast institution-level topic trends with uncertainty.	[4], [5], [7]
G2	Quality assessment relies on lagging citations or manual rubrics; automated scoring must be explicit and auditable.	[6], [8]
G3	Early-warning prediction for research projects is rare outside learning analytics and often opaque.	[9], [10]

1.5 Aim and Objectives

Aim. To design, implement and evaluate a Research Performance Analytics and Visualization Module that forecasts research-domain trends, scores research quality,

predicts project success and visualises the results for data-driven decisions by students, supervisors and administrators.

- **SO1** — Implement time-series forecasting of research-domain activity with a target accuracy of at least 78% (1 − MAPE).
- **SO2** — Develop a four-dimension quality score with weights originality 30%, citation impact 25%, methodology 25% and clarity 20%.
- **SO3** — Build interactive visualisation for trends and scores.
- **SO4** — Build a success-prediction model (target F1 above 0.75 and ROC-AUC above 0.80) for early identification of at-risk work.
- **SO5** — Integrate the module with the other modules through REST APIs and shared tables.

1.6 Research Questions

- **RQ1:** How accurately do per-topic ARIMA models forecast publication activity in the SLIIT corpus, compared with a naïve baseline?
- **RQ2:** Can a blend of trained models and calibrated heuristics produce quality and success scores that discriminate genuine research writing from non-research text, and what do the reported training metrics actually measure?

1.7 Scope, Limitations and Assumptions

In scope: yearly publication counts per broad discipline from the SLIIT corpus; title-and-abstract quality and success scoring; PDF/TXT upload. **Out of scope:** citation-count prediction, full-text peer review, and individual staff performance evaluation. **Assumptions:** the corpus is a reasonable proxy for institutional research activity; abstracts reflect a paper's methodology and contribution; the success definition (Section 4.4.3) is a pragmatic proxy, not a ground truth.

1.8 Individual Contribution

The author designed and implemented the Module 4 service: `trend_forecaster.py` (518 lines), `quality_predictor.py` (407), `success_predictor.py` (203), `topic_classifier.py`, the routers

`trends.py`, `quality.py`, `papers.py` and `prediction.py`, the training pipeline (`prepare_training_data.py`, `train_trend_forecaster.py`, `train_quality_model.py`, `train_success_predictor.py`, `train_topic_classifier.py`), migration `005_module4_tables.sql`, and the three analytics pages including the 857-line trends dashboard.

1.9 Thesis Structure

Chapter 2 reviews related work; Chapter 3 the methodology; Chapter 4 requirements and design; Chapter 5 implementation, integration and testing; Chapter 6 results; Chapter 7 discussion; Chapter 8 commercialisation; and Chapter 9 conclusions, following the SLIIT guidelines [11].

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

Searches in IEEE Xplore, ACM Digital Library, Scopus-indexed journals and Google Scholar (2006–2026) used *research trend forecasting*, *ARIMA*, *Prophet*, *bibliometric mapping*, *research quality indicators*, *gradient boosting* and *data leakage*. Foundational statistical texts were included for time-series methods.

2.2 Domain and Theoretical Foundations

2.2.1 Time-series forecasting

ARIMA(p,d,q) models combine autoregression, differencing and moving-average terms and produce prediction intervals from the fitted residual variance [12]. Hyndman and Athanasopoulos emphasise evaluation on hold-out data and comparison with simple benchmarks such as the naïve forecast [13]. Prophet decomposes a series into trend, seasonality and holidays and tolerates missing data [14]. Makridakis et al. found that simple statistical methods often match or outperform machine-learning methods on short univariate series [15]. Annual publication counts per topic are short, noisy series with structural breaks, which favours parsimonious models and honest benchmarking.

2.2.2 Research trend analysis

VOSviewer and CiteSpace map co-citation and keyword networks and detect bursts [4], [5]; Asooja et al. forecast emerging trends from scientific literature using term-frequency time series [7]. These tools target the global literature; few systems forecast within one institution's repository.

2.2.3 Research quality assessment

Aksnes et al. explain that citations capture scholarly impact rather than quality as such and accumulate slowly [6]; Thelwall investigated whether large language models can evaluate research quality and found only moderate agreement with expert scores [8]. For student work, transparent rubric-like scores computed from observable features are more defensible than opaque judgements.

2.2.4 Gradient boosting and data leakage

XGBoost is a scalable, regularised gradient-boosting system widely used for tabular prediction [9]; SHAP explains its predictions feature by feature [10]. Kaufman et al. define leakage as the use of information in training that would not be available, or that trivially encodes the target, at prediction time; leakage inflates evaluation metrics without improving real performance [16].

2.2.5 Interactive visualisation

D3 provides interactive, data-driven visualisation for the web [17], and bibliometric tools show that visual exploration helps users interpret research landscapes [4].

2.3 Related Research

Table 2.1: Related research informing Module 4

Study	Approach	Relevance and limitation
Box et al. [12]	ARIMA methodology	Basis of the forecaster; needs adequate series length.
Taylor and Letham [14]	Prophet	Planned ensemble partner; not deployed.
Makridakis et al. [15]	Forecasting competition analysis	Motivates naïve-baseline comparison.
Asooja et al. [7]	Trend forecasting from literature	Global literature; term-based.
van Eck and Waltman [4]	Bibliometric mapping	Descriptive, not predictive.
Aksnes et al. [6]	Citations and quality	Citation lag unsuitable for student work.
Chen and Guestrin [9]	Gradient boosting	Quality regressors and success classifier.
Kaufman et al. [16]	Leakage in data mining	Explains near-perfect training metrics.

2.4 Existing Systems, Products and Techniques

Table 2.2: Comparison of existing products with Module 4

Product	Trend forecast	Quality score	Success prediction	Institution-specific
VOSviewer / CiteSpace	Burst detection	No	No	If data supplied
Scopus / SciVal analytics	Descriptive trends	Citation metrics	No	Subscription
Power BI dashboards	Manual	Manual	No	Yes
Learning-analytics early warning	No	No	Course-level	Yes

Product	Trend forecast	Quality score	Success prediction	Institution-specific
Module 4	ARIMA with 95% CI	4-dimension, explainable	Blended probability + risk	SLIIT corpus

2.5 Gap Synthesis and Conceptual Framework

The review suggests that institutional analytics should (i) forecast from the institution's own output with uncertainty and benchmark against simple baselines, (ii) score quality on explicit, weighted dimensions, (iii) treat predictive models with care for leakage and blend them with transparent rules, and (iv) present results interactively. Figure 2.1 places these services in the student journey.

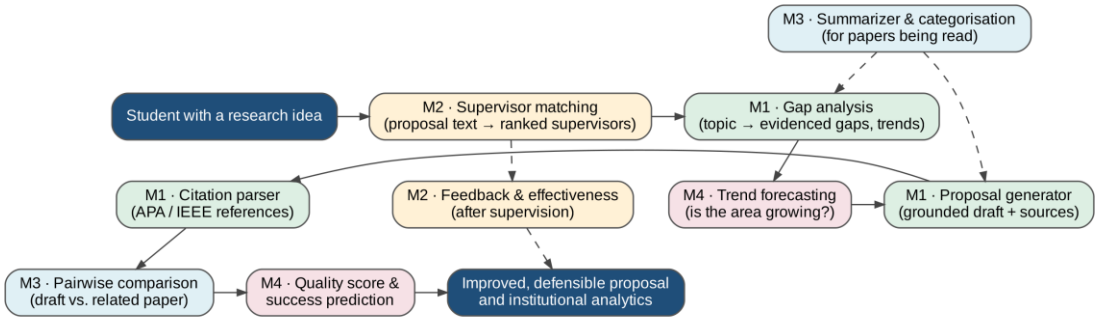


Figure 2.1: Student journey through the integrated platform; Module 4 informs topic choice (trends) and quality checking before submission

3 RESEARCH METHODOLOGY

3.1 Research Design

The work follows Design Science Research [18], [19]: forecasting, scoring, prediction and visualisation artefacts were built in iterations and evaluated with in-sample and hold-out metrics, an independent sample of corpus abstracts and functional validation through the running service.

3.2 Data, Participants and Study Materials

Table 3.1: Data and study materials used by Module 4

Material	Description	Use
SLIIT corpus	4,219 records (3,860 with usable abstracts after filtering) from Module 3	All models
Trend data	Yearly counts per topic: general, all, business, social sciences, computing, health, engineering, sciences (4–27 points each)	ARIMA forecaster
Quality / success data	3,860 papers with 11 extracted features; 3,281 train / 579 test	XGBoost regressors and classifier
Topic data	1,533 labelled abstracts, six disciplines; 1,303 train / 230 test	SBERT + logistic regression
Evaluation sample	600 randomly sampled corpus abstracts (≥ 200 characters, seed 42)	Score distributions and latency
Test texts	Strong technical abstract, HR leave form, empty and too-short inputs	Functional validation

3.3 Experimental Procedure

1. Prepare yearly topic counts and per-paper features from the corpus (`prepare_training_data.py`).
2. Fit ARIMA(1,1,1) per topic and record in-sample RMSE (`train_trend_forecaster.py`).
3. Train XGBoost multi-target regressors for the four quality dimensions and overall score, and an XGBoost success classifier, on an 85/15 split.
4. Train the SBERT + logistic-regression discipline classifier.

5. Back-test the forecaster: refit ARIMA(1,1,1) on years up to 2023 and forecast 2024–2025, comparing with the naïve last-value forecast (excluding the partial 2026 year).
6. Score 600 sampled corpus abstracts with the deployed quality and success services and record distributions and latency.
7. Execute the 15-case functional validation on 28 August 2026.

3.4 Measures and Evaluation Metrics

Table 3.2: Evaluation measures for Module 4

Objective	Measure	Target
SO1	In-sample RMSE, MAE, NRMSE; hold-out MAPE versus naïve; 1 − MAPE	1 − MAPE ≥ 78%
SO2	Model R^2 on test split; discrimination between research and non-research text; score distribution on 600 abstracts	Explainable, discriminative
SO3	Functional rendering of charts, comparisons, insights and reports	Functional
SO4	Accuracy, ROC-AUC; discrimination on validation texts	F1 > 0.75, AUC > 0.80
SO5	Scores available to other modules through REST APIs	Functional

3.5 Data Analysis

$$\text{MAPE} = (100/n)\,\Sigma\,|y_t - \hat{y}_t| \,/\, \max(y_t, 1); \quad \text{NRMSE} = \text{RMSE} / \text{mean}(y)$$

Hold-out errors were computed on two years per topic and interpreted cautiously given the series lengths. Score distributions were summarised by mean, standard deviation and histogram. Validation outcomes follow the PASS/PARTIAL/FAIL/BLOCKED scheme.

3.6 Ethics, Privacy, Security and Responsible AI

- The corpus consists only of publicly available bibliographic records and abstracts from SLIIT's open research repository; full texts uploaded by users are stored in a private Supabase Storage bucket protected by Row-Level Security [20].
- Personal data (names, e-mail addresses, research interests, ratings) are processed in line with the principles of Sri Lanka's Personal Data Protection

Act No. 9 of 2022 — purpose limitation, data minimisation and access control [21].

- Every request to an intelligence service passes through the gateway, which verifies a Supabase-issued JSON Web Token [22] and applies rate limiting (600 requests per minute globally and 200 per minute for ML routes).
- Generative components (Gemini) are used only where interpretation is genuinely required and every retrieval-based output is traceable to a cited SLIIT paper, which reduces the risk of hallucinated content [23].
- Quality and success scores are presented as formative feedback with recommendations, not as grades, and expose their dimension scores and features.
- A research-document check warns when a non-research text is scored, preventing misleading numbers.

3.7 Reproducibility and Research Artefacts

Training scripts, model pickles with metadata (`models/trained_*/metadata.json`), trend data and the validation harness (`docs/module-reports/_test_module4.py`) are versioned in the repository; the back-test and 600-abstract evaluation are reproducible from the stored models and corpus with a fixed random seed.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1: Functional requirements of Module 4

ID	Requirement
FR1	Forecast yearly publication counts per topic for a chosen horizon with 95% intervals, direction, growth and accuracy.
FR2	Compare multiple topics and identify emerging topics and recommended research areas; export an HTML report.
FR3	Score a title and abstract (or PDF/TXT) on four quality dimensions with an overall score, predicted discipline and recommendations.
FR4	Predict success probability with label, confidence, risk band and recommendations.

Table 4.2: Non-functional requirements of Module 4

ID	Category	Requirement
NFR1	Performance	Scoring under 1 s; forecasts from pre-fitted models.
NFR2	Explainability	Dimension scores, weights, features and model versions returned.
NFR3	Fairness	Scores must not penalise legitimate technical vocabulary.
NFR4	Reliability	Each model degrades independently to a heuristic.
NFR5	Usability	Interactive charts with confidence bands; SUS above 70 (proposal target).

4.2 Component Architecture within the Complete Solution

Figure 4.1 shows the component. Every router is mounted under the shared `/analytics` prefix; the gateway route file `module4.routes.ts` applies authentication and rate limiting with a 60-second timeout. A recurring design pattern is that every trained model is blended with a calibrated heuristic rather than trusted alone, because the training targets were derived from narrow heuristics that can be unfair to legitimate technical papers.

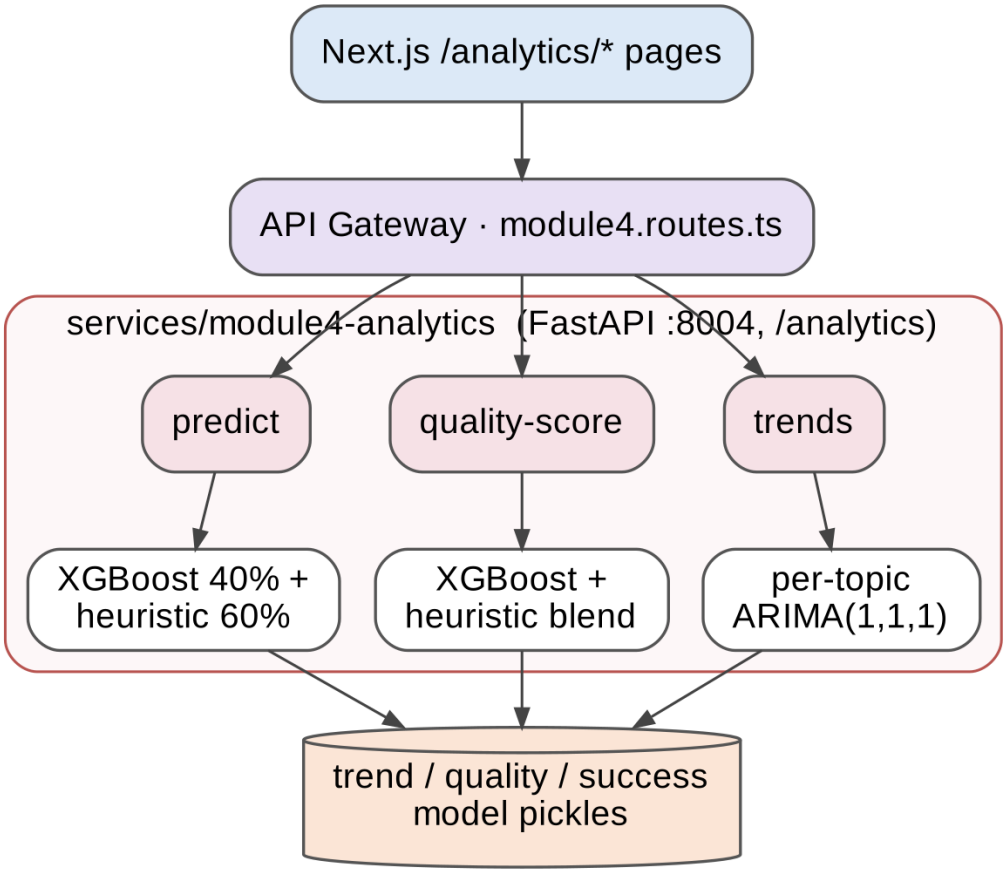


Figure 4.1: Component architecture of the Research Performance Analytics and Visualization Module

4.3 Design Alternatives and Decisions

Table 4.3: Design decisions and rationale for Module 4

Concern	Alternative considered	Decision	Rationale
Forecasting	ARIMA–Prophet ensemble weighted by MAPE [14]	Per-topic ARIMA(1,1,1)	Short annual series; principled intervals [12]; fewer moving parts.
Quality scoring	Model only / heuristic only	Per-dimension blend of XGBoost and calibrated heuristic	Model learned a narrow heuristic; blend restores fairness to CS papers.
Success prediction	Random Forest + XGBoost soft voting	XGBoost 40% + heuristic 60%	Avoids extreme 0%/100% outputs; explainable.
Discipline label	TF-IDF classifier (Module 3)	SBERT + logistic regression (6 classes)	Broader disciplines; better generalisation.

4.4 Detailed Design

4.4.1 Trend forecasting

Figure 4.2 shows the forecaster. Each topic's fitted `statsmodels` ARIMA results object [24] is loaded from `trend_models.pkl` with its historical years and counts. `get_forecast(steps=h)` returns the mean forecast and the 95% prediction interval; negative values are clamped to zero and the upper bound is capped at three times the historical maximum to avoid pathologically wide bands. Accuracy is computed against in-sample fitted values, aligned for differencing. Direction, growth, emergence and recommendation are:

direction: mean(last 3) > 1.2 × mean(earlier) → rising; < 0.8 × → declining; otherwise stable

$$\text{growth\%} = (\text{forecast_end} - \text{latest_historical}) / \text{latest_historical} \times 100$$

$$\text{emergence} = 0.5 \cdot (\text{slope_last4} - \text{slope_all}) + 0.5 \cdot \text{latest} / \text{mean}(\text{counts}[:-3])$$

$$\text{recommendation} = \text{clamp}(\text{growth\%}/100, -1, 2) - 0.4 \cdot \text{latest} / \text{max_latest}$$

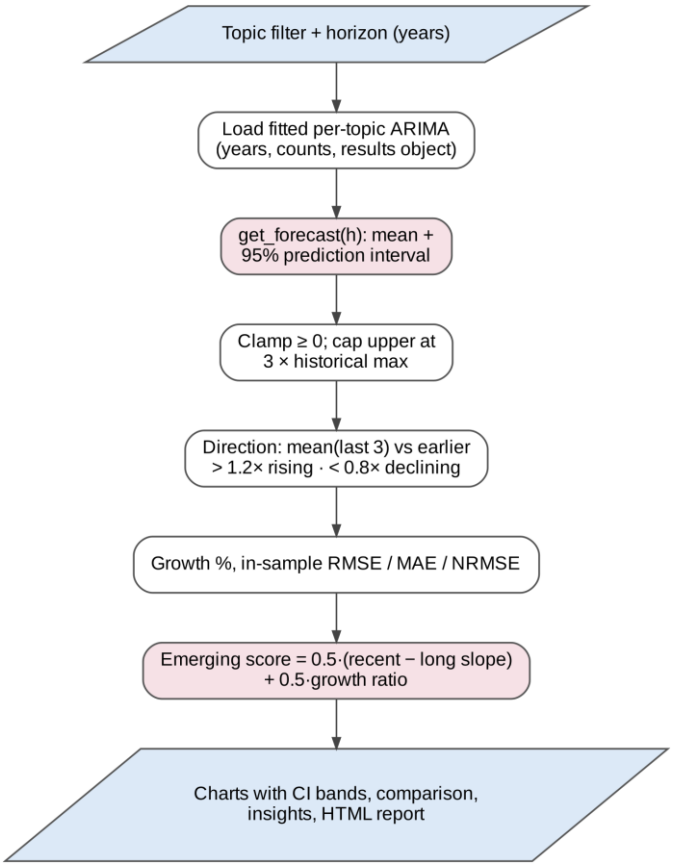


Figure 4.2: Per-topic ARIMA forecasting with prediction intervals, direction, growth and emergence

4.4.2 Quality scoring

`extract_features()` computes 11 features: word, title-word and sentence counts, average word and sentence length, methodology-keyword count (25 terms), author count, citation signals (bracketed numbers, author–year parentheses, "et al."), year, abstract length and title length. A calibrated heuristic computes each dimension using research-phrase detection and 45 strict methodology terms:

$$\text{originality} = \min(1,\ 0.30 \cdot \min(1,\ |\text{abstract}|/1200) + 0.70 \cdot \text{research_signal})$$

$$\text{methodology} = \min(1,\ \text{strict_method_terms} / 4); \quad \text{clarity} = \text{clamp}(1 - \max(0,(\bar{w} - 7.5)/4) - \max(0,(\bar{s} - 30)/30),\ 0.30,\ 1)$$

where $\bar{w}$ and $\bar{s}$ are average word and sentence lengths. When the XGBoost bundle is available, each dimension is blended with its model prediction at a dimension-specific ratio (model : heuristic) — methodology 0.10 : 0.90, originality 0.15 : 0.85, citation impact 0.20 : 0.80 and clarity 0.40 : 0.60 — and the overall score is recomputed:

overall = 0.30 · originality + 0.25 · citation_impact + 0.25 · methodology + 0.20 · clarity

`is_research_document()` checks for academic sections and phrases and attaches a warning when a non-research text is scored (Figure 4.3).

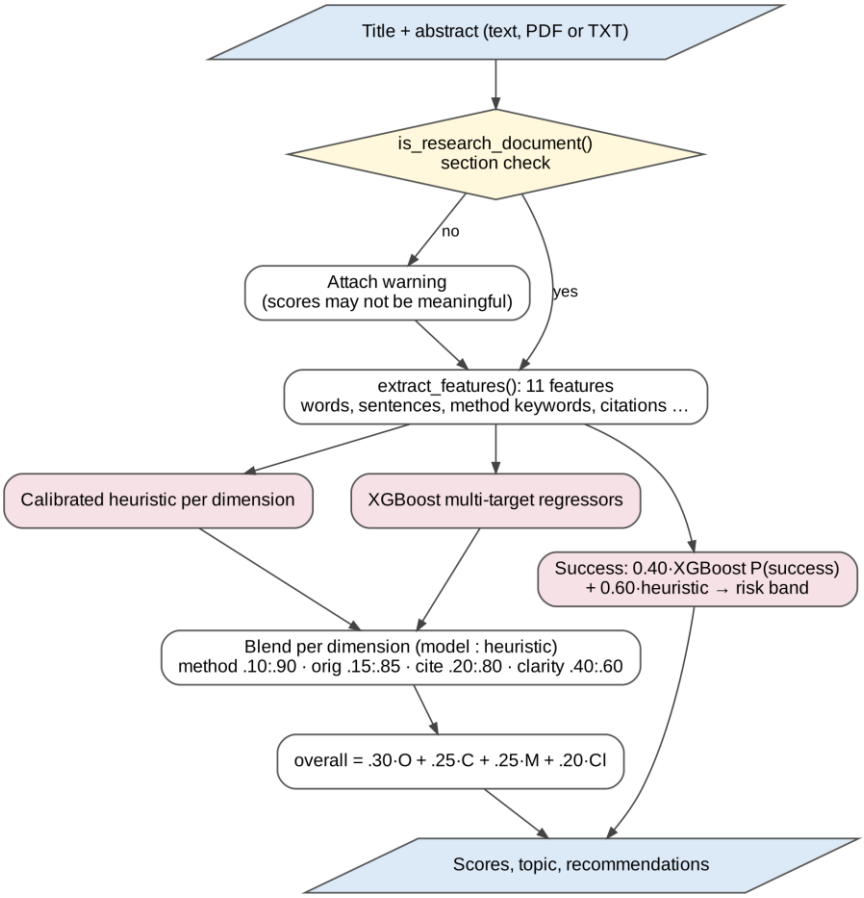


Figure 4.3: Quality scoring and success prediction: feature extraction, heuristic and model blend

4.4.3 Success prediction

The success classifier reuses the same 11 features. Training labels were defined as overall quality ≥ 0.62 *and* abstract length ≥ 500 characters *and* at least two methodology keywords. At inference:

$$P(\text{success}) = 0.40 \cdot \text{XGBoost.predict_proba}(x) + 0.60 \cdot \text{heuristic}(x)$$

The label is *successful* if $P \geq 0.5$; risk is low for $P \geq 0.7$, medium for $P \geq 0.4$ and high otherwise. Recommendations are generated from the features that fell short (for example fewer than three methodology keywords or fewer than two citation signals).

PDF uploads are parsed with PyMuPDF, skipping header lines that match DOI/copyright patterns and locating the abstract marker.

4.5 Standards and Professional Practice

- Forecast evaluation follows hold-out and benchmark practice [13]; model reporting includes version identifiers in every response.
- Quality attributes follow ISO/IEC 25010 [25]; web security follows OWASP guidance [26].
- Accessibility of charts (labels, contrast, tooltips) follows WCAG 2.1 principles [27].

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1: Technology stack

Layer	Technologies	Justification
Presentation	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, shadcn/ui, Recharts, D3 / react-force-graph-2d, Zustand, Axios	Server-side rendering, typed components, responsive layout and rich interactive charts in one framework [28]
Integration	Express.js API gateway (TypeScript), express-rate-limit, Supabase JWT verification, per-module proxy with 60 s / 150 s timeouts	Single entry point that authenticates once and isolates the front end from service locations [29]
Intelligence	Python 3.10+, FastAPI, sentence-transformers (all-MiniLM-L6-v2, fine-tuned), scikit-learn, XGBoost, statsmodels, PyMuPDF / pypdf / pdfplumber, Google Gemini 2.5 Flash client	Asynchronous typed micro-services; CPU-friendly models that fit the hosting budget [30], [31]
Data	Supabase PostgreSQL 15 + pgvector, Row-Level Security, Supabase Auth and Storage; pre-computed NumPy embedding matrices and pickled model bundles	Relational and vector search in one managed service, policy-based access control [32], [20]
DevOps	pnpm workspaces + Turborepo monorepo, Docker multi-stage images, GitHub Actions, Vercel, Railway, Azure Container Registry workflow, Playwright	Reproducible builds and automatic deployment on every push to main [33]

Module-specific dependencies include `statsmodels` (ARIMA), `xgboost`, `scikit-learn`, `sentence-transformers`, `PyMuPDF` and Recharts on the front end.

5.2 Component Implementation

Table 5.2: Key Module 4 implementation files

File	Lines	Responsibility
app/services/trend_forecaster.py	518	Forecasts, intervals, accuracy, emergence, recommendations, report
app/services/quality_predictor.py	407	Features, research check, heuristic, model blend
training/prepare_training_data.py	324	Topic counts and per-paper features from the corpus
app/routers/papers.py	208	Analyse text and PDF/TXT upload
app/routers/trends.py	205	Trends, compare, insights, topics, report
app/services/success_predictor.py	203	Success probability blend, risk, recommendations
training/train_quality_model.py	169	XGBoost multi-target regressors

File	Lines	Responsibility
training/train_success_predictor.py	159	XGBoost success classifier
web: analytics/trends/page.tsx	857	Forecast charts with CI, comparison, insights, report
web: analytics/quality-scores/page.tsx	371	Quality scoring UI with upload and paste modes
web: analytics/predictions/page.tsx	316	Success prediction UI

The success probability blend is a single, auditable expression in `success_predictor.py`:

```
heuristic = _heuristic_score(features)
...
model_proba = float(model.predict_proba(X)[0][1])
# Blend: heuristic dominates (60 %) so technical papers are not unfairly
# penalised by the model's over-fitted narrow training distribution.
proba = round(0.40 * model_proba + 0.60 * heuristic, 4)
```

Figures 5.1 to 5.3 show the implemented pages.

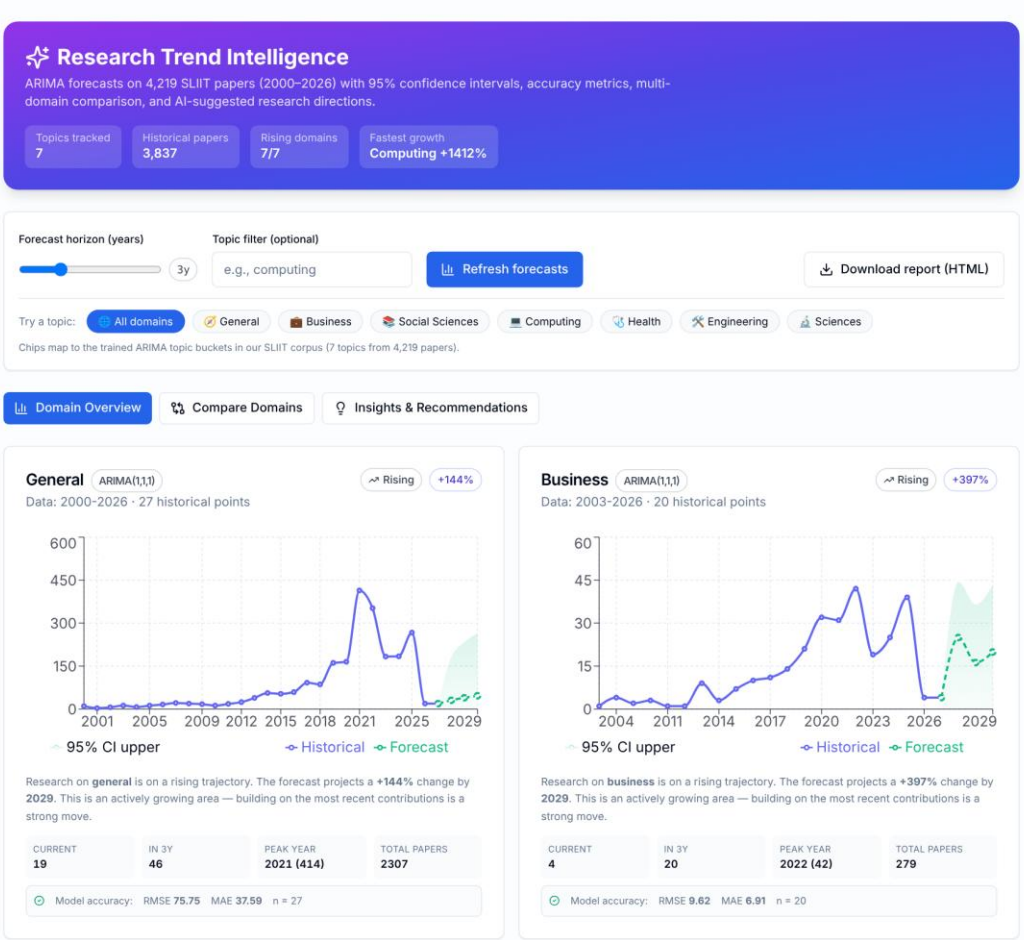


Figure 5.1: Research Trend Intelligence page: forecast controls and per-topic ARIMA charts with 95% confidence bands and accuracy

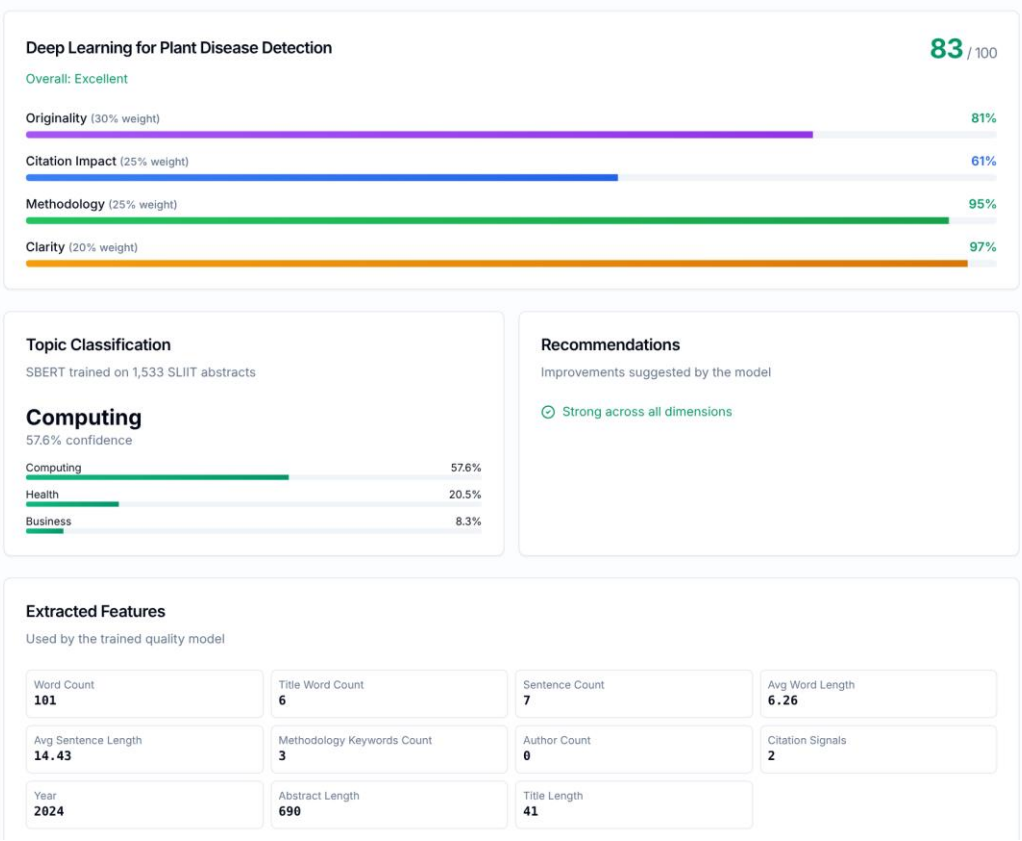


Figure 5.2: Quality scoring result for a technical abstract: overall 83/100 with dimension scores, discipline prediction and extracted features

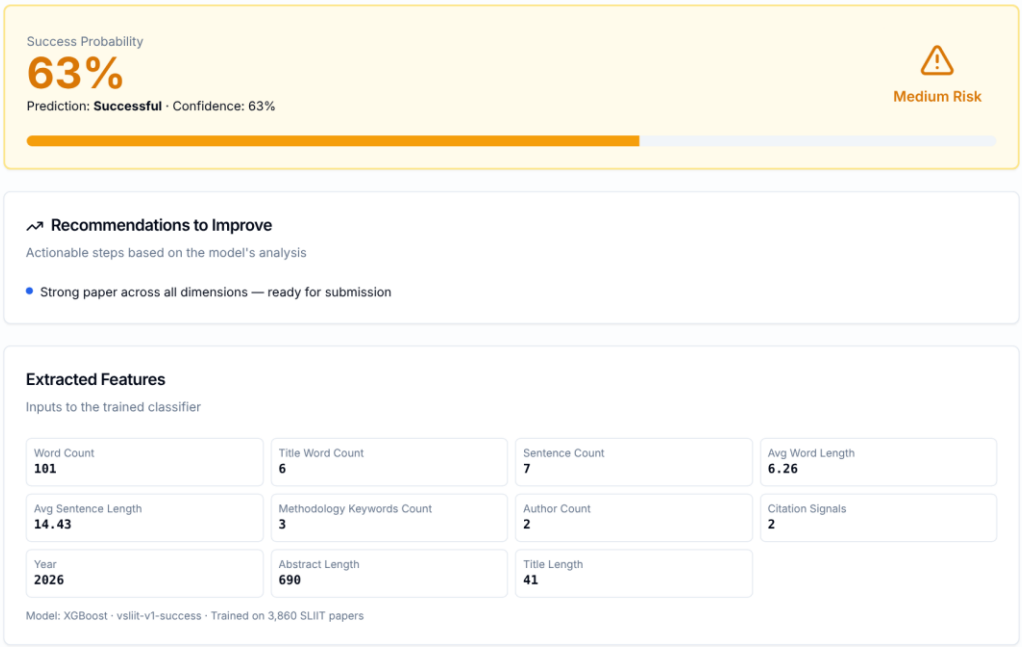


Figure 5.3: Success prediction result: probability 63%, prediction successful, medium risk, with extracted features

5.3 Integration with the Other Components

- **Module 3:** all models are trained on the corpus maintained by Module 3.
- **Module 1:** generated proposals can be scored for quality and success; trend forecasts complement gap analysis.
- **Platform:** shared gateway, authentication and dashboard shell (Figure 5.4).

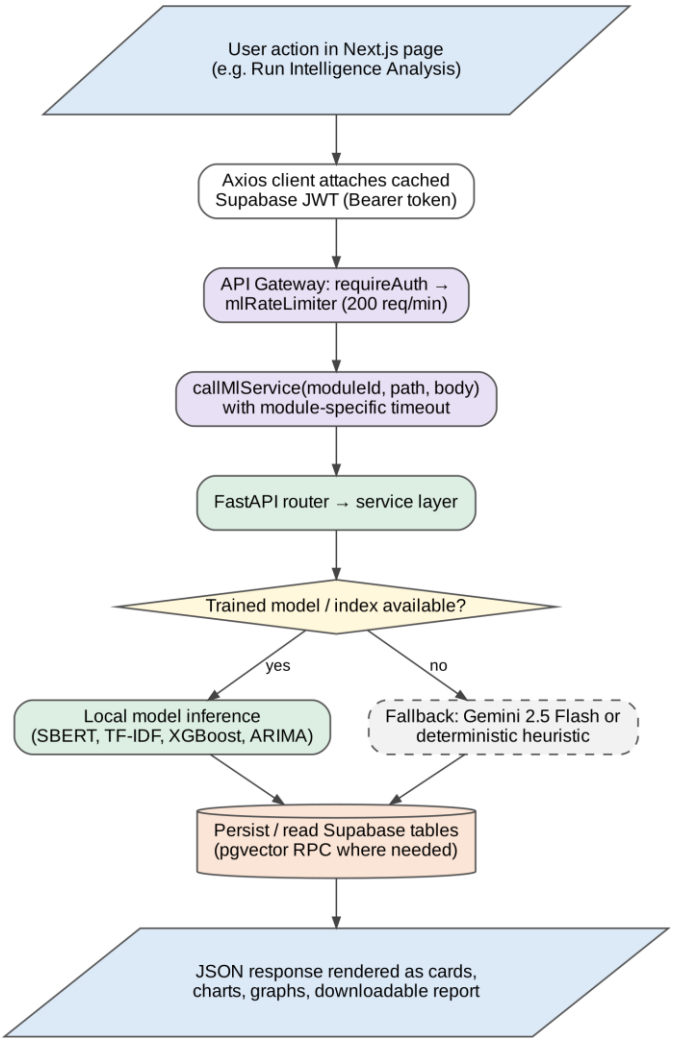


Figure 5.4: Request flow from a Next.js page through the gateway to a FastAPI service

5.4 Verification and Software Testing

Model-level evaluation is reported in Chapter 6. Functional validation executed 12 cases against the running service on 28 August 2026 (Table 5.3); the Playwright test `e2e/module4-trends.spec.ts` covers the trends page.

Table 5.3: Functional validation of Module 4 (28 August 2026)

ID	Scenario	Result	Evidence or root cause
M4-01	Trends — all topics, horizon 3	FAIL	All topics "rising", growth +80% to +1,412%: baseline is partial 2026 count
M4-02	Trends — computing, horizon 5	FAIL	Same growth artefact
M4-03	Trend comparison	FAIL	Ranking driven by artefact
M4-04	Trend insights	FAIL	Template text contradicts negative emergence score
M4-05	Available topics	PASS	Eight fitted topics listed
M4-06	Quality — strong technical abstract	PASS	overall 0.828 (O 0.81, C 0.61, M 0.95, Cl 0.97)
M4-07	Quality — HR leave form	PARTIAL	overall 0.236 (correctly low) but clarity high
M4-08	Quality — missing input	PASS	HTTP 400 with clear message
M4-09	Success — strong abstract	PASS	P = 0.632, successful, medium risk; blend hand-verified
M4-10	Success — leave form	PASS	P = 0.200, needs improvement
M4-11	Success — too-short abstract	PASS	HTTP 400 (minimum 50 characters)
M4-15	Health and model metadata	PASS	Four bundles loaded with recorded metrics

5.5 Deployment and Operation

The service is containerised with the model pickles and SBERT baked into the image and deployed on Railway with automatic redeployment (Figure 5.5); PyMuPDF was added for PDF analysis and PDF processing time was reduced on 13 May 2026.

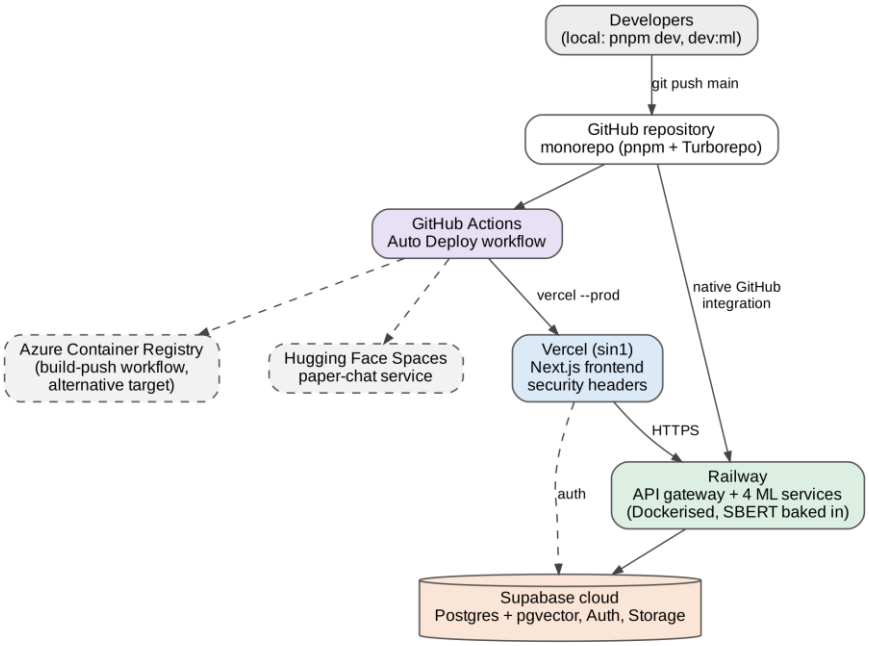


Figure 5.5: Continuous deployment pipeline of the platform

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 RQ1: Forecast accuracy

Figure 6.1 reports in-sample fit for each topic. Errors scale with series size: RMSE ranges from 1.2 (sciences, four points) to 75.8 papers per year (general), with normalised RMSE between 0.69 and 1.23, indicating that one-step in-sample errors are comparable to the mean annual count for several topics because of the 2021 spike and the partial 2026 year.

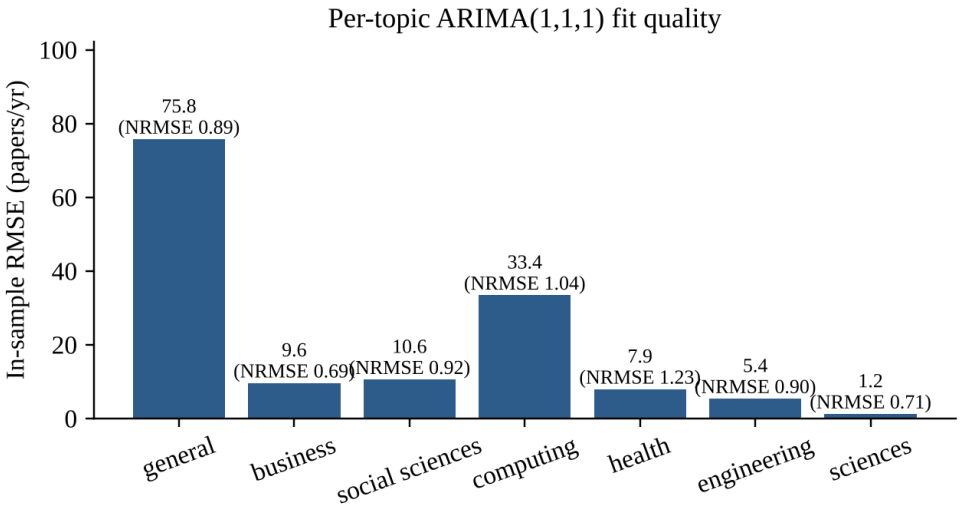


Figure 6.1: In-sample RMSE and normalised RMSE of per-topic ARIMA(1,1,1) models

The hold-out back-test (training up to 2023, forecasting 2024–2025) is summarised in Figure 6.2 and Table 6.1. ARIMA achieved its best accuracy on the largest series (general: MAPE 15.3%, i.e. 1 − MAPE = 84.7%, above the 78% target) and narrowly beat the naïve forecast for general and business, but was worse than naïve for computing, health, engineering and social sciences. For computing the model forecast near-zero counts while the actual counts were 53 and 144, illustrating the sensitivity of differenced models to the structural break after 2022.

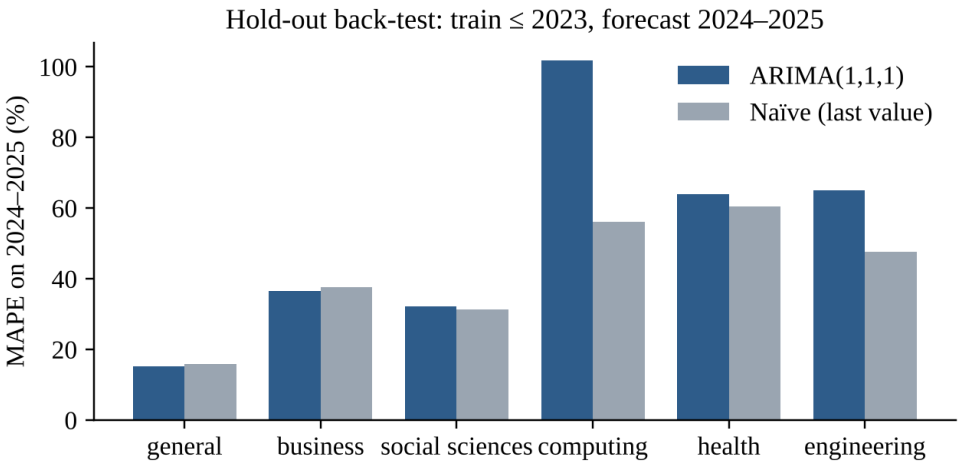


Figure 6.2: Hold-out MAPE of ARIMA(1,1,1) versus a naïve last-value forecast

Table 6.1: Hold-out back-test, 2024–2025 (actual versus forecast)

Topic	Actual 2024, 2025	ARIMA forecast	ARIMA MAPE	Naïve MAPE	1 − MAPE
General	184, 267	188, 190	15.3%	16.0%	84.7%
Business	25, 39	32, 21	36.5%	37.6%	63.5%
Social sciences	20, 45	19, 19	32.3%	31.4%	67.7%
Computing	53, 144	0, 0 (clamped)	101.8%	56.1%	—
Health	6, 32	4, 4	63.9%	60.4%	36.1%
Engineering	9, 16	20, 17	65.1%	47.6%	34.9%

Validation also revealed that the headline growth percentage divided the forecast by the partial-year 2026 count (for example computing: 4 papers in 2026 versus 144 in 2025), so every topic was labelled rising with growth between +80% and +1,412%. RQ1 is answered: per-topic ARIMA is adequate only for the largest, smoothest series and does not reliably beat a naïve baseline; incomplete final years must be excluded.

6.1.2 RQ2: Quality and success scoring

The XGBoost quality regressors reported test R^2 between 0.997 and 0.999 (overall MAE 0.0034) and the success classifier accuracy 0.983 and ROC-AUC 0.9994 on 579 test papers, nominally exceeding the F1 0.75 and AUC 0.80 targets. These figures must be interpreted with care: the regression targets were produced by a heuristic computed from the same 11 features, and the success label was defined by thresholds on overall quality, abstract length and methodology keywords — all functions of the input features. The models therefore learned to reproduce the labelling rule, a form of target

leakage [16], and the metrics do not measure agreement with expert judgement. This is why the deployed scores are dominated by calibrated heuristics.

On validation texts, scoring discriminated correctly: a strong technical abstract scored 0.828 overall and success probability 0.632, while an HR leave form scored 0.236 and 0.200, and invalid inputs were rejected. On 600 randomly sampled corpus abstracts (Figure 6.3), the mean overall quality was 0.495 (SD 0.115); clarity averaged 0.92 while citation impact (0.32) and methodology (0.33) were low because abstracts rarely contain citations or explicit method terms. Success prediction labelled 157 abstracts (26%) successful, with 142 low-, 206 medium- and 252 high-risk cases (Figure 6.4). Median scoring latency was 1.6 ms for quality and 0.4 ms for success.

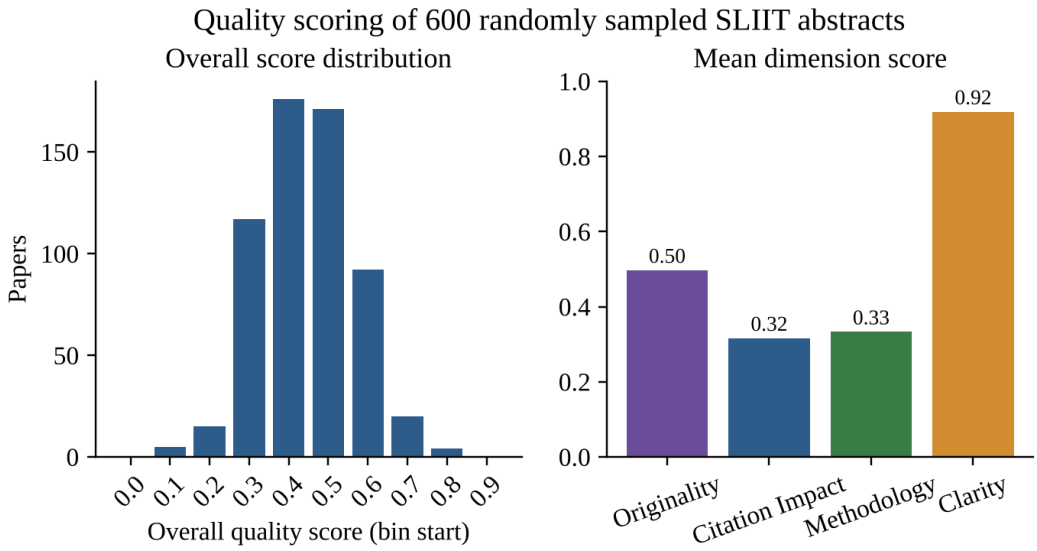


Figure 6.3: Quality scores of 600 sampled SLIIT abstracts: overall distribution and mean dimension scores

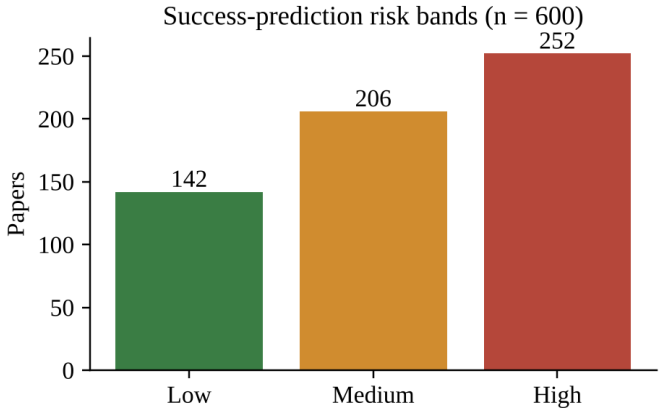


Figure 6.4: Success-prediction risk bands for 600 sampled abstracts

The six-class discipline classifier used for quality reports reached accuracy 0.709, macro-F1 0.627 and weighted F1 0.714 on 230 test abstracts, and classified the technical validation abstract as computing (0.58).

6.2 Product and Technical Results

Table 6.2: Feature completion status of Module 4

Feature	Status	Notes
Trend forecasting with 95% intervals	Complete	Growth baseline defect (partial year)
Comparison, insights, recommendations, report	Complete	Insight text template defect
Quality scoring (text, PDF, TXT)	Complete	Heuristic-dominated blend
Success prediction	Complete	Heuristic-dominated blend
Discipline classifier	Complete	Accuracy 0.71
Prophet ensemble; Random Forest	Not deployed	Planned in proposal

6.3 User and Stakeholder Validation

The supervisor reviewed and signed off the module validation report, highlighting quality scoring and success prediction as well validated and the growth calculation as the priority. A task-based user-acceptance protocol including the System Usability Scale is provided in Appendix D.

6.4 Integration Results

All eight Module 4 routes are reachable through the gateway, and quality scores and predictions can be stored in Supabase for later review.

6.5 Summary of Findings

Quality scoring and success prediction work as transparent, fast, discriminative feedback tools, but their high training metrics reflect leakage rather than validity; ARIMA forecasting is useful for large series but needs baseline benchmarking and exclusion of partial years. In validation, 7 of 12 cases passed, 1 was partial and 4 failed (Figure 6.5).

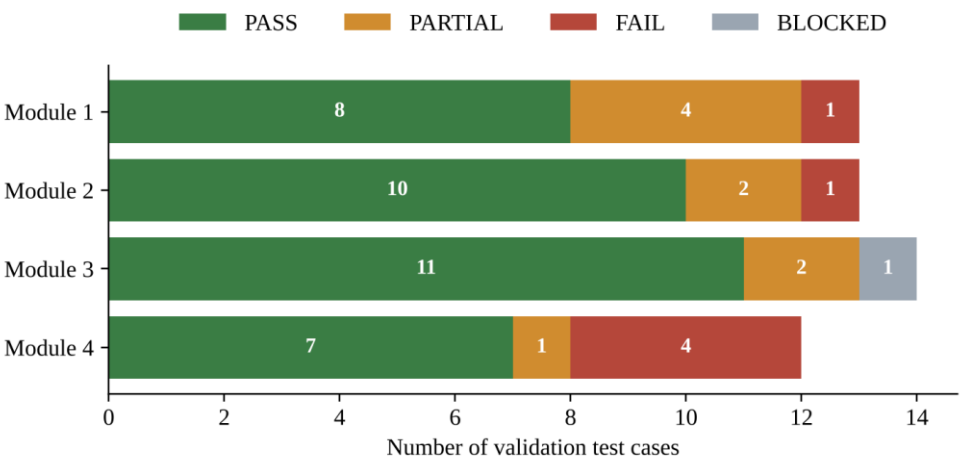


Figure 6.5: Validation outcomes per module (Module 4 in the bottom row)

7 DISCUSSION

7.1 Interpretation of Findings

The results show a consistent theme: analytics are only as good as the time frame, labels and dependencies behind them. The growth defect and the computing back-test failure both arise from data recency (a partial final year and a post-2022 shift), not from ARIMA itself; this agrees with forecasting literature that urges hold-out evaluation against simple baselines [13], [15]. The near-perfect quality and success metrics illustrate textbook leakage [16]; the decision to let calibrated heuristics dominate the blend was therefore sound, because the heuristics are explicit and their behaviour on legitimate technical papers was checked by hand. Low citation-impact and methodology scores on corpus abstracts are expected, since abstracts seldom contain citations — these dimensions are more informative on full proposals.

7.2 Comparison with Literature Baselines and Existing Solutions

Bibliometric tools describe trends [4], [5]; Module 4 adds institution-level forecasts with uncertainty, but its accuracy is comparable to a naïve forecast for most topics, so its added value lies mainly in intervals, visual comparison and recommendations. Compared with LLM-based quality judgements, which show only moderate agreement with experts [8], the scores here are explicit and reproducible but are not yet validated against expert ratings (the proposal target of Cohen's $\kappa \geq 0.70$ [34] remains to be tested).

7.3 Research Contribution

- An institution-level forecasting service with prediction intervals, emergence scoring and recommendations, together with an honest hold-out comparison against a naïve baseline.
- A transparent, per-dimension blending scheme of gradient-boosted models and calibrated heuristics for formative quality feedback, with an explicit analysis of target leakage.
- Evidence on engineering pre-conditions (dependencies, training status, data recency) for advanced analytics in student research platforms.

7.4 Product Contribution

Students get immediate, explainable feedback on drafts and a view of which areas are active; supervisors get early warning of weak work. All responses carry model versions and features, supporting accountability.

7.5 Limitations and Threats to Validity

- **Forecasting:** short series, a partial 2026 year, a growth-baseline defect and a template insight text; only two hold-out years per topic.
- **Leakage:** quality and success training metrics reflect reproduction of a heuristic; no expert-rated ground truth.
- **Sample:** the 600-abstract evaluation scores abstracts, not full proposals.
- **Planned components:** Prophet ensemble, Random Forest and Power BI integration were not delivered.

7.6 Implications

Institutions should treat automated scores as formative signals and invest in expert-labelled data before using them for decisions. Forecasting services should always display a naïve benchmark and exclude incomplete periods. Teams deploying advanced models should include dependency and model-registry checks in their release process.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

Primary users are supervisors, department heads and research administrators (approximately 30–55 years old) who oversee student research portfolios; **secondary users** are students seeking feedback and topic guidance. The module replaces a combination of bibliometric tools, spreadsheets, manual rubric scoring and ad-hoc risk tracking with a single analytics view.

8.2 Market and Competitive Position

Commercial research-analytics suites are subscription-based and citation-centric; bibliometric tools are descriptive; learning-analytics early-warning systems operate at course level (Table 2.2). The module's position is institution-specific, formative analytics integrated with the research workflow, priced for South Asian universities.

8.3 Business and Sustainability Model

Table 8.1: Indicative revenue model (from the approved proposal)

Phase	Strategy	Indicative pricing
Year 1	Free for SLIIT departments; data collection and model refinement	—
Year 2	Freemium: trend views free; quality scoring and prediction premium	LKR 400 per student per month
Year 3	Institutional licence with custom analytics and support	LKR 200,000 per institution per year

The proposal estimated LKR 150,000 per year for hosting and a development budget of LKR 125,000; CPU-only inference (sub-millisecond scoring) keeps marginal costs low.

8.4 Scalability, Intellectual Property and Compliance

- Forecasts are pre-fitted; scoring is O(1) per request and results can be cached.
- The blending scheme, heuristic calibration, emergence and recommendation scores are the team's original work; libraries are used under open-source licences.

- Scores relating to individuals are personal data under the PDPA [21]; they must not be used for summative decisions without validation, consent and appeal mechanisms.

8.5 Contribution to the Sustainable Development Goals

The platform contributes primarily to **SDG 4 (Quality Education)**, in particular target 4.3 (equal access to quality tertiary education) and target 4.4 (relevant skills), by giving every student — not only those with well-connected supervisors or paid tools — structured support for research. It also contributes to **SDG 9 (Industry, Innovation and Infrastructure)**, target 9.5 (enhance scientific research and upgrade technological capabilities), by turning an institutional repository into queryable research infrastructure [35]. Module 4 contributes through formative feedback that helps students improve before assessment (SDG 4) and through institutional research analytics that inform curriculum and capacity planning (SDG 9).

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Research Performance Analytics and Visualization Module of Researchly AI. It delivered interactive trend forecasts with uncertainty, transparent quality scoring and success prediction, and it evaluated them honestly: forecasts are useful mainly for large topics and must exclude partial years; and high training metrics for scoring reflect leakage, which the heuristic-dominated blend mitigates.

9.2 Achievement of Objectives

Table 9.1: Achievement of objectives

Objective	Status	Evidence
SO1 Forecasting (1 − MAPE ≥ 78%)	Partly achieved (84.7% for general; lower elsewhere)	Section 6.1.1; Table 6.1
SO2 Four-dimension quality score	Achieved (expert agreement untested)	Section 6.1.2
SO3 Interactive visualisation	Achieved	Section 5.2; Figures 5.1–5.3
SO4 Success prediction (F1 > 0.75, AUC > 0.80)	Achieved on proxy labels; validity limited by leakage	Section 6.1.2
SO5 Integration	Achieved	Section 6.4

9.3 Individual Research and Product Contributions

- Built the forecasting service with intervals, emergence and recommendations, and back-tested it against a naïve baseline.
- Designed the feature set, calibrated heuristics and per-dimension model blend for quality and success scoring.
- Implemented the analytics interfaces for trends, quality scoring and success prediction.

9.4 Future Directions

- Exclude incomplete years and compute growth from the last complete year; fix insight templates; add naïve and Prophet benchmarks with automatic model selection [14].
- Collect expert ratings for 100–200 proposals, measure Cohen's κ [34] and retrain models on expert labels; add SHAP explanations [10].
- Run the SUS-based user study in Appendix D.

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GLOSSARY

Term	Definition
Prediction interval	A range expected to contain a future observation with a stated probability (here 95%).
Naïve forecast	A benchmark that predicts the last observed value for all future periods.
Target leakage	Training information that trivially encodes the label, inflating evaluation metrics.
Calibrated heuristic	An explicit rule-based score tuned by inspection to behave sensibly on known cases.
Emergence score	A measure of how much a topic's recent slope exceeds its long-term slope.
Risk band	Low, medium or high category derived from success probability thresholds.

APPENDICES

Appendix A: Extended Results and Test Evidence

Table A.1: Recorded training metrics of Module 4 models

Model	Training / test size	Reported metrics
Quality regressors (XGBoost)	3,281 / 579	R^2: originality 0.9993, citation 0.9983, methodology 0.9994, clarity 0.9972, overall 0.9971
Success classifier (XGBoost)	3,281 / 579	Accuracy 0.9827, ROC-AUC 0.9994
Discipline classifier (SBERT + LogReg)	1,303 / 230	Accuracy 0.709, macro-F1 0.627, weighted F1 0.714
ARIMA(1,1,1) per topic	4–27 yearly points	RMSE 1.2–131.8 papers/year (Figure 6.1)

Recorded response excerpt for validation case M4-06:


```
{ "overall_score": 0.828, "originality_score": 0.8098,
  "citation_impact_score": 0.6118, "methodology_score": 0.9491,
  "clarity_score": 0.9741, "topic": { "primary_topic": "computing",
"confidence": 0.5757 } }
```

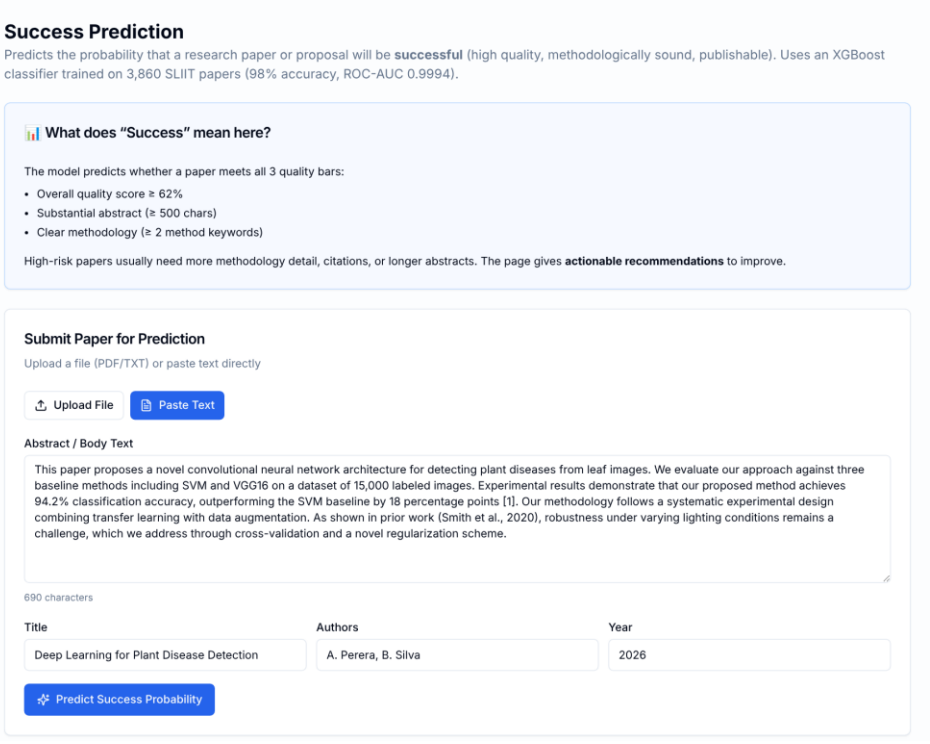



Figure A.1: Success Prediction input page with the definition of success

Appendix B: Academic Integrity and AI-Use Disclosure

Table B.1: AI-use disclosure

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic) — AI coding assistant	Debugging suggestions; execution of the module validation harness and drafting of the module report (28 Aug 2026)	Evidence inspected; supervisor sign-off obtained	Chapter 5.4, Chapter 6
Claude (Anthropic)	Drafting and formatting this thesis; running the back-test and 600-abstract evaluation from stored models; diagrams and charts	Scripts re-run; numbers cross-checked with model metadata	All chapters

Appendix C: Additional Interface Screenshots

Screenshots were captured from the production build of the Researchly AI front end rendering responses recorded from the live Module 4 service during the 28 August 2026 validation run.

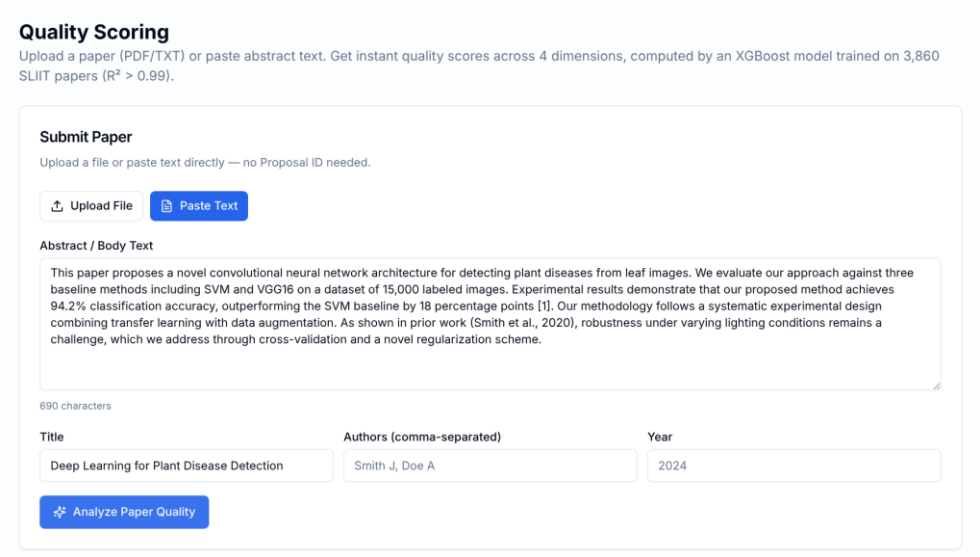


Figure C.1: Quality Scoring input in paste-text mode

Appendix D: User Acceptance Testing Protocol

Participants complete the following tasks unaided; completion, time and errors are recorded and the System Usability Scale [36] is administered afterwards (target above 70).

Table D.1: UAT tasks and recording sheet for Module 4

Task	Expected outcome	Completed (Y/N)	Time (s)	Errors / comments
T1 Identify whether "computing" research is growing and read its 95% interval	Correct reading			
T2 Compare three topics and download the trend report	Report saved			
T3 Score your abstract and name its weakest dimension	Dimension identified			
T4 Predict success for your proposal and list two recommendations	Recommendations read			